Run report: cen-q7-split-run-20261024T2239Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part CENSUS (sampled) under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: cen-q7-split 20261024T2239Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 efce43f3fdaa848389a03f3d778a9b9604fe23d1d6e5e1bd75e777a49fb6cf01
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 26415a19d8d07de211c7992a6c97ca99a2a3a358fabb5c02e7893639121efd5b
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/cen-q7-split-run-20261024T2239Z CENSUS 7 split 30000000 202610247 0 2000
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 93
```

tool_log.txt

```
split q=7: samples=30000000 (random) res_shaped=18820 allowed_trace_candidates=827 smooth_prime_order_hits=827 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=1356/2000 secs=93.8
split q=7: total prime-order mass estimate 676.228 +- 23.515 (hit rate per sampled form 2.757e-5)
```

summary.txt

```
split q=7: samples=30000000 (random) res_shaped=18820 allowed_trace_candidates=827 smooth_prime_order_hits=827 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=1356/2000 secs=93.8
split q=7: total prime-order mass estimate 676.228 +- 23.515 (hit rate per sampled form 2.757e-5)
```

cand_t_hist.csv (first 12 of 187 rows)

type	t	candidates
split	−98	21
split	−97	1
split	−96	4
split	−95	4
split	−94	15
split	−93	1
split	−92	10
split	−91	8
split	−90	23
split	−89	13
split	−88	27
split	−87	6

hits.csv (first 12 of 827 rows)

type	code	t	nE	smooth
split	21465600853535	−45	2447	true
split	8256776209770	69	2333	true
split	13932065985012	55	2347	true
split	18817065944093	−45	2447	true

split	9615597712178	9	2393	true
split	22379408582650	3	2399	true
split	12103450054528	51	2351	true
split	8392089724110	−57	2459	true
split	2357605659177	−45	2447	true
split	10798782165558	45	2357	true
split	28838902113595	9	2393	true
split	12688304214622	61	2341	true

mass_by_t.csv (first 12 of 19 rows)

type	t	nE	smooth_hits	mass_estimate	mass_stderr
split	−75	2477	43	35.1606	5.3619
split	−71	2473	34	27.8014	4.7679
split	−65	2467	44	35.9783	5.4239
split	−57	2459	42	34.3429	5.2992
split	−45	2447	59	48.2436	6.2808
split	−39	2441	21	17.1714	3.7471
split	−15	2417	37	30.2545	4.9738
split	−9	2411	45	36.7960	5.4852
split	3	2399	68	55.6028	6.7428
split	9	2393	67	54.7851	6.6931
split	13	2389	65	53.1497	6.5924
split	19	2383	82	67.0504	7.4045

Graph

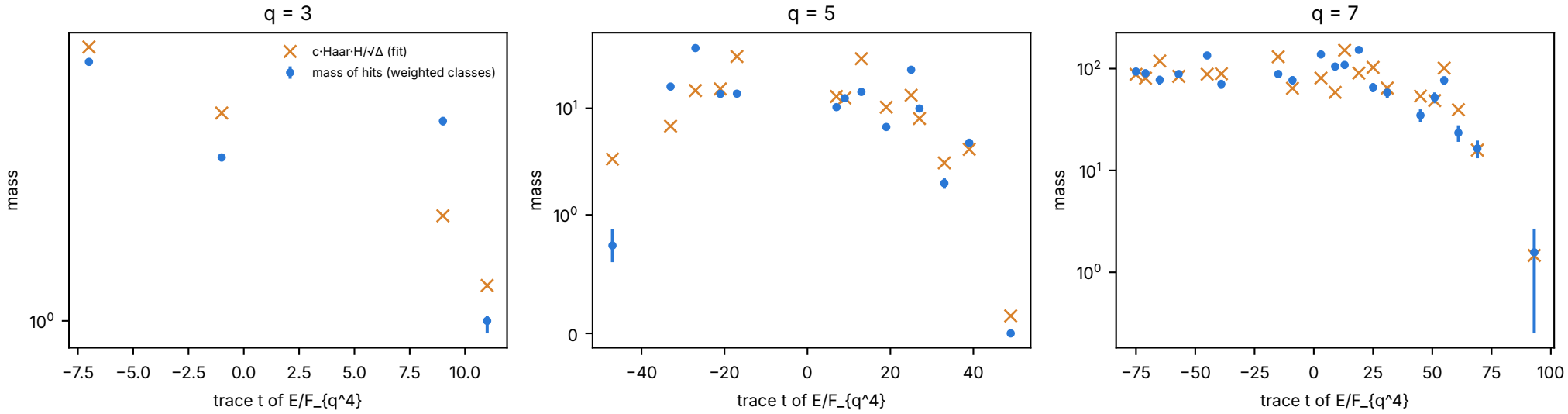


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.